

URL shortner

Functional requirements

user should be able to shorten long URL into short URL,

shortened URL should be unique,

End user should be redirected to appropriate webpage once he clicked on it

Non-functional requirements:

data-partitioning,

load balancing,

fault tolerance,

vertical scalability,

horizontal scalability,

low latency,

high throughput,

CAP theorem (Here eventual consistency is fine , but here we need to prioritise 99.99% uptime
i.e Availability >> Consistency)

Core Entities:

user,

url,

history

API:

POST:/shorten

{

userID,

originalURL

}

GET:/click/{url}

HLD:

